

Projective Incidence Calculus for Transformer Decision Logic

J. Allan Scott
Independent Researcher
Concord, New Hampshire, USA
allan@jallan.scot

ABSTRACT

We introduce *Projective Incidence Calculus* (PIC), a geometric formalism that models transformer decision logic as signed evidence accumulation in a Hilbert space, and use it as a principled *diagnostic lens* on real models. PIC turns the additive read-out into measurable structure. Applying it to direct-logit-attribution dumps from the fieldrun probing infrastructure on *two model families*—Qwen2.5 (0.5B–3B) and Pythia (70M–1B)—we find that routing is strongly *rank-concentrated* in both: the decode rides a stable, late-layer, MLP-dominated circuit using far fewer effective blocks than are available. *Coherence packing, by contrast, is family-dependent*. Qwen routing features sit 2–3× above the information-theoretic (Welch) floor, whereas Pythia packs near-optimally (1.06–1.5×); at matched dimension ($n_b=49$), Pythia-410M (1.38×) is markedly tighter than Qwen2.5-0.5B (2.01×), showing that near-Welch-optimal packing is achievable in general pretraining and is *not* limited to task-specialized regimes. These are structural properties of routing invisible to accuracy or loss curves. The formalism rests on a machine-checked backbone: PIC modernizes classical incidence calculus (set-based incidences become inner products, structural overlap a Gram kernel) and connects the hard ($T=0$) decision surface to tropical geometry via Laguerre power diagrams. On this base we define *Tropical Realization Complexity* (TRC), the cost of routing inputs into the correct cells, and prove its core results: exact softmax/decode recovery, a decode-separation lemma with a γ -code packing bound, and a generator-rank bound. On a controlled synthetic substrate, four theory-guided interventions all fail to outperform plain gradient descent. These results are clarifying: they show that the framework’s value is diagnostic rather than algorithmic, because (i) the capacity bound is slack by ~ 50 –59 orders of magnitude, (ii) realization cost is dominated by the number of trainable generators, not the number of routing decisions—a *superposition* quantity—and (iii) gradient descent already packs routing features near the Welch floor when given a well-defined task. We release the fieldrun infrastructure, the i-orca machine-checked certificate corpus, and the pil learning code.

CCS CONCEPTS

• Computing methodologies → Machine learning; • Theory of computation → Logic.

KEYWORDS

transformers, tropical geometry, incidence calculus, mechanistic interpretability, routing, superposition, formal verification

1 INTRODUCTION

A decoder transformer [18] predicts the next token by iteratively writing additive contributions into a residual stream and reading

the result through an unembedding matrix. This additive structure is well understood at the level of implementation: every layer, attention head, and MLP block adds a vector, and the logits are a linear read-out of the sum. What has been missing is a unified *geometric and logical* account of how those contributions combine into a *confident decision*—and a precise statement of what limits the confident decisions a given read-out geometry can support.

We propose such an account. *Projective Incidence Calculus* (PIC) treats each residual contribution as casting a vector of signed *incidences* (inner products) onto a frame of proposition directions, the unembedding rows. The decode is the tropical (max-plus) collapse of the accumulated incidences, and its geometry is a Laguerre power diagram. PIC recovers the softmax exactly at temperature $T=1$ and the greedy decode at $T=0$; the two live in one program under two semirings. On this base we define *Tropical Realization Complexity* (TRC), a measure of the cost of routing a data distribution into the correct cells.

The payoff of this geometry is *measurement*. PIC names quantities—rank concentration, the read-out circuit, coherence relative to the Welch floor, difficulty-driven fallback—that a practitioner can read off a frozen model and that behave lawfully across scale, language, and difficulty (§6). *While PIC and TRC do not yield superior training algorithms in the regimes we tested, they provide a principled diagnostic lens that reveals structural properties of real-model routing invisible to standard accuracy or loss curves*. That is the paper’s main claim, and it is what the rest of the contributions support.

To establish where the framework measures rather than optimizes, we use a learning system, *Projective Incidence Learning* (PIL), which owns and updates the proposition frame and a bank of generated rules. On controlled substrates with a *known* ground-truth factorization, PIL lets us ask which parts of the theory are levers for training and which are only good for measurement. The answer is sharp—and the clarifying negative results (§5.5) are precisely what make the positive real-model findings interpretable.

Throughout we separate three epistemic statuses and tag every claim accordingly, so that the formal and empirical content never blur: [PROVED] (machine-checked end-to-end, with no admitted steps, in the companion i-orca corpus [15]), [MEASURED] (empirical, this work or fieldrun), and [OPEN] (conjecture). The paper’s arc is measure-first: define the diagnostics, validate them on two real model families, and use a controlled synthetic substrate to show *why* they are diagnostics rather than training levers. Concretely, our contributions are:

- **A formal geometric model of transformer decision logic** (§3) grounded in incidence calculus and tropical geometry, with three machine-checked structural results: exact softmax/decode recovery, a decode *separation* lemma yielding a γ -code packing bound on confidently decodable tokens, and a generator-rank bound.

- **Tropical Realization Complexity** (§4) and the empirical finding (§5.4) that it is a *superposition-capacity* quantity: realization cost scales with the number of trainable generators, not the number of routing decisions, and the binding constraint in practice is per-example routing, not cell capacity (which is slack by ~ 50 – 59 orders).
- **A set of clarifying controls** (§5): four theory-guided interventions—frame regularization, min-margin rule targeting, propose-score-select, and a generator coherence penalty—each fail to beat plain gradient descent, because SGD already packs routing features near the Welch floor. This delimits where the theory is, and is not, an algorithm.
- **A two-family real-model validation** (§6) over Qwen2.5 (0.5B–3B) [13] and Pythia (70M–1B) [3], plus a difficulty dial and (in the appendix) five languages: rank concentration on a stable late-MLP circuit transfers to both families, while coherence packing is *family-dependent*—Qwen sits 2 – $3\times$ above the Welch floor and Pythia packs near it, refuting the prior reading that near-optimal packing is synthetic-specific.
- **Open-source tooling**: the `fieldrun --pil-dump` [14] direct-logit-attribution seam, the *i-orca* certificate corpus [15], and the `pil` learning code [16], with every figure regenerable from checked-in result files.

2 BACKGROUND

2.1 Transformers and direct logit attribution

A transformer maintains a residual stream that is the running sum of additive contributions: an embedding, and, per layer, one attention output and one MLP output. Writing U for the unembedding, the final logit of token v is $L_v = \langle r, U_v \rangle + b_v$ where r is the final residual. Because $r = \sum_j d_j$ is a sum, $L_v = \sum_j \langle d_j, U_v \rangle + b_v$ decomposes *exactly* into per-block contributions. This is *direct logit attribution* (DLA): the contribution $\langle d_j, U_v \rangle$ of block j to token v [8, 12]. We take the additive blocks ($n_b = 2L+1$ for an L -layer model) as the atomic “circuits” and their per-candidate contributions as the incidences.

2.2 Incidence calculus

Classical incidence calculus [5, 9] attaches to each proposition the *set of possible worlds (incidences)* in which it holds, and computes with these sets rather than with scalar probabilities alone. This preserves the dependencies that a purely numeric calculus discards and handles non-truth-functional combination faithfully. PIC keeps the spirit—propositions carry structured *evidence* that combines, not just a number—but replaces the Boolean lattice of world-sets with the inner-product geometry of a Hilbert space, so that “incidence” becomes a signed real (a DLA term) and “structural overlap” becomes a Gram entry.

2.3 Tropical geometry and neural networks

Zhang, Naitzat, and Lim [20] showed that feedforward ReLU networks are tropical rational maps: decision boundaries are tropical hypersurfaces and linear regions correspond to vertices of an associated Newton polytope. This has been extended to real tropical geometry [4], decision-boundary analysis [2], and linear-region

Table 1: The machine-checked backbone of PIC and TRC (all [PROVED], with no admitted steps, in the *i-orca* corpus [15]). Empirical claims are tagged [MEASURED] in situ; conjectures [OPEN].

result	one-line statement
Softmax recovery (Thm. 1)	product-of-experts over incidences = softmax at $T=1$, greedy decode at $T=0$
Decode separation (Thm. 2)	two γ -decodable tokens have frames $\geq \gamma$ apart; biases cancel
γ -code packing (Cor. 1)	at most $(1 + 2\rho/\gamma)^d$ confidently decodable tokens fit in $\mathbb{R}^d$
Generator rank (Thm. 3)	M generators move the logits within a $\leq \min(M, d)$ -dim. subspace
Routing Welch (Thm. 4)	$n > m$ routing features in $\mathbb{R}^m$ force coherence $\geq$ the Welch floor

counting [11, 17]. We apply these ideas not to a feedforward classifier but to the *routing/decode* behavior of a transformer read-out, where the relevant tropical object is the upper envelope of the affine logit functions—a Laguerre power diagram [10].

3 PROJECTIVE INCIDENCE CALCULUS

PIC models the transformer read-out in a Hilbert space $\mathcal{H} = \mathbb{R}^d$. Sources (circuits / partial evidence) are vectors $d_j \in \mathcal{H}$; the proposition frame $U = \{U_v\}_{v \in V} \subset \mathcal{H}$ is the set of unembedding rows. Table 1 lists the formal backbone established in this section and the next; it is the load-bearing distinction of the paper, so we state it once, up front—these results are [PROVED], everything in §5–6 is [MEASURED].

Definition 1 (Incidence). The incidence of source d_j on proposition U_v is the inner product $c_j^v = \langle d_j, U_v \rangle$ (the DLA term).

Definition 2 (Aggregate logits). With $r = \sum_j d_j$, the logit of v is $L_v = \langle r, U_v \rangle + b_v = \sum_j c_j^v + b_v$.

Definition 3 (Gram kernel). Structural overlap between propositions is the Gram matrix $G_{vw} = \langle U_v, U_w \rangle$. High $|G_{vw}|$ is the near-synonym / common-mode regime in which differential evidence collapses; a diagonal G is the classical (Boolean-incidence) limit.

Two algebras. The model factors into an *inner, linear* algebra—accumulation $r = \sum_j d_j$ and attribution $c_j^v = \langle d_j, U_v \rangle$ are entirely linear/bilinear—and an *outer, tropical* algebra: the decode $\hat{v} = \arg \max_v L_v$ is a max-plus collapse. The first result makes the bridge exact.

THEOREM 1 (SOFTMAX/DECODE RECOVERY; [PROVED], *i-orca*: RecoveredProbability). *The product-of-experts model whose per-source likelihoods are the exponentiated incidences returns exactly the transformer softmax at temperature $T=1$, and the greedy $\arg \max$ decode at $T=0$. The same program is sound under both the log-semiring and the tropical (max-plus) semiring (a Maslov sandwich).*

The tropical decision surface. At $T=0$ the hard decision $\arg \max_v L_v$ is governed by the tropical polynomial $M(r) = \max_v (\langle r, U_v \rangle + b_v)$. Its regions are the cells of a *Laguerre power diagram* with additive

weights $\|U_v\|^2 - 2b_v$: the decision region of v is $C_v = \{r : L_v \geq L_w \forall w\}$, and the normalized decode margin is the (signed) distance from r to the nearest cell facet. This is the transformer-routing instance of the ReLU-as-tropical-rational picture [10, 20].

How many confident decodes fit? The central structural question is: given a frame in $\mathbb{R}^d$, how many tokens can be decoded with a guaranteed margin? Call v γ -decodable if some unit residual ($\|r\| \leq 1$) decodes to v with margin $\geq \gamma$. The answer is a clean separation lemma whose proof cancels the biases.

THEOREM 2 (DECODE SEPARATION; [PROVED], i-orca: DecodeCapacity). *If $v \neq w$ are both γ -decodable, then $\|U_v - U_w\| \geq \gamma$, independent of the biases b .*

PROOF SKETCH. Let r_v, r_w be the unit witnesses. Margin $\geq \gamma$ at r_v gives $\langle r_v, U_v - U_w \rangle + (b_v - b_w) \geq \gamma$, and symmetrically $\langle r_w, U_w - U_v \rangle + (b_w - b_v) \geq \gamma$. Adding cancels the biases: $\langle r_v - r_w, U_v - U_w \rangle \geq 2\gamma$. Cauchy-Schwarz with $\|r_v - r_w\| \leq 2$ yields $\|U_v - U_w\| \geq \gamma$. $\square$

COROLLARY 1 (γ -CODE PACKING BOUND). *The γ -decodable set is a γ -separated code in $\mathbb{R}^d$, so by sphere packing $N_\gamma \leq (1 + \frac{2\rho}{\gamma})^d$, with $\rho = \max_v \|U_v\|$.*

This is the decision-side sibling of the Welch bound: Welch [19] bounds the *coherence* of V vectors in $\mathbb{R}^d$; Corollary 1 bounds the *count* of γ -separated decodes. Both are packing statements. A companion HeadTail certificate shows that a compact “head” of propositions reproduces the full decode *iff* it tropically dominates the remaining “tail,” and that the certifiable head is itself capacity-bounded by Corollary 1—the tail is the explicit irreducible residue (the “forge tax” of open-class tokens).

Adjustments are low-rank. Finally, when the read-out is augmented by M trainable generators (rules), the set of achievable logit adjustments is constrained.

THEOREM 3 (GENERATOR RANK; [PROVED], i-orca: RoutingRank). *The logit adjustments realizable by M generators lie in a subspace of dimension at most $\min(M, d)$. When the number of routing features to be distinguished exceeds this rank, superposition is forced.*

4 TROPICAL REALIZATION COMPLEXITY

Definition 4 (Tropical Realization Complexity). The TRC of a data distribution $\mathcal{D}$ at margin γ and interference tolerance ϵ is the minimal number M of trainable generators that route $\mathcal{D}$ into the correct power-diagram cells with margin $\geq \gamma$ and routing-feature interference $\leq \epsilon$.

TRC decomposes into two components with very different epistemic status.

Rank component ([PROVED]). By Theorem 3, M generators move the logits within a $\leq \min(M, d)$ -dimensional subspace. If a task requires distinguishing more routing features than this rank, they must share dimensions—superposition is unavoidable. This is the dimension budget of TRC.

Interference component ([MEASURED]/[OPEN]). Within the available rank, packing n routing features incurs mutual coherence. The structural floor is provable: for n unit features in m dimensions with $n > m$, mean-squared coherence is $\geq (n - m)/(m(n - 1))$.

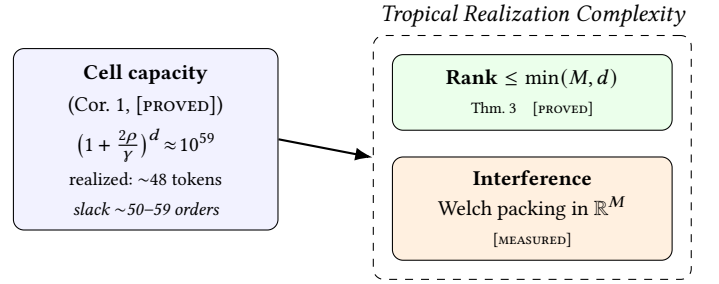


Figure 1: Cell capacity vs. Tropical Realization Complexity. The γ -code packing bound permits $\sim 10^{59}$ confident decodes but only ~ 48 are realized—capacity is not the binding limit. TRC is: a [PROVED] rank budget ($\leq \min(M, d)$) plus a [MEASURED] interference term that is a Welch packing of routing features into $\mathbb{R}^M$ (rule-activation space), set in practice by per-example routing.

THEOREM 4 (ROUTING WELCH FLOOR; [PROVED], i-orca: RoutingWelch). *$n > m$ unit routing features in $\mathbb{R}^m$ have mean-squared coherence at least the Welch bound $(n - m)/(m(n - 1))$; equality requires an equiangular tight frame.*

We deliberately do *not* promote the floor to a margin-degradation law. Our measurements (§5.4) show the coherence $\rightarrow$ margin link is empirically mild; asserting a clean $\gamma \leq \gamma_0(1 - c\mu)$ theorem would contradict the data.

Empirical correction (§5.4). The packing dimension for the interference term is M —the rule-activation space—not $\min(M, d)$. This is the single most important refinement to Definition 4: TRC is a Welch packing of n routing features into M dimensions, and interference appears precisely when $n > M$.

5 EXPERIMENTS: SYNTHETIC SUBSTRATE

We construct synthetic problems with a *known* ground-truth factorization so that “which lever matters” has a checkable answer. The model owns a proposition frame U and (optionally) a bank of ReLU rules $h_k(z) = \text{relu}(\langle w_k, z \rangle + b_k)$ emitted as extra sources. Every loss term is labelled *decode-side* (the NLL and a worst-competitor hinge margin—exactly what fieldrun measures on a frozen model) or *frame-side* (the frame potential, mean-squared off-diagonal cosine, driven toward the Welch floor). Unless noted, evaluation is on a held-out 25% split and results are averaged over 2–3 seeds.

The section answers one question—*which lever moves the decode?*—in three moves. §5.1–5.2 contrast the two candidate levers (frame geometry vs. generation), finding only generation works. §5.3–5.4 then show *why* the theory’s refinements of generation do not help: capacity does not bind, and realization cost is an M -dominated superposition quantity. §5.5 collects the resulting four nulls, the section’s highest-signal takeaway.

5.1 Frame optimization is decoupled from the decode

We first test the most natural “geometric” lever: regularize the frame to be more incoherent. We use an 8 $\times$ over-complete regime ($d=24$, $V=192$, 24 planted synonym clusters, within-cluster $|\cos|=0.82$,

Table 2: Frame regularization is decoupled from the decode (hard over-complete regime, $d=24$, $V=192$, 3 seeds). The frame potential moves; the decode does not.

frame_reg	retr	top1	margin	nll	fp/Welch
0.00	0.395	0.99	1.240	1.949	1.346
0.10	0.400	0.99	1.241	1.949	1.244
0.50	0.395	0.99	1.242	1.948	1.094

Welch floor 0.037). It behaves as designed: synonymy compresses margins so hard that retrievability is 0.40 while top-1 accuracy is 0.99, and the strongest competitor is a synonym 88% of the time. Sweeping the frame-potential weight *does* move the frame geometry (frame potential $1.35 \rightarrow 1.09 \times$ Welch) but leaves *every* decode metric flat (Table 2). Across the entire synonymy $\times$ noise plane, $|\Delta \text{retr}| \leq 0.01$ while the baseline retrievability swings $0.02 \rightarrow 0.80$ under the real levers (synonymy spread and source SNR). The frame regularizer is *redundant* where the margin term already shapes the frame and *impotent* where the bottleneck is source distinguishability: when synonyms have near-identical sources, the margin is source-SNR-bound, and no read-out objective can manufacture absent signal.

5.2 Generation is the lever

§5.1 says the bottleneck is generative, not the frame. We test this directly with a compositional regime: synonyms share a one-hot *topic* atom (so they collide on r), and a fraction of clusters are *hard*—the synonym is the XOR of two code atoms, provably not linearly separable—while the rest are *easy* (parity read directly). A frame-only model is a linear read-out and is pinned at chance on the hard clusters (within-synonym accuracy 0.454, margin ≈ 0). Adding generated ReLU rules lifts hard-cluster accuracy to 0.93 (Fig. 2). A signal=False control (parity is noise) collapses all arms to chance, confirming the honest boundary: generation recovers signal only when it exists.

But the second half is a null. *Min-margin-targeted* rule placement—seeding rules at the at-risk facets discovered from a frame-only warmup—barely beats *random* rules (0.934 vs. 0.911 at budget 32; behind at low budget). The mechanism is clear: gradient descent already drives hidden units to the hard clusters, so explicit targeting buys initialization, not allocation.

5.3 Selection is not the lever; capacity does not bind

To separate “proposal quality” from “trained vs. frozen feature,” we cross a selection method (random, variance, η^2 -on-ambiguous-examples) with frozen vs. trainable input weights (Table 3). The split is *entirely* along frozen-vs-trainable: making the scored arm trainable jumps it from 0.53 to 0.74 at $M=8$, matching SGD, while every frozen arm stalls. Once rules are trained, theory-guided selection is indistinguishable from random initialization; once frozen, no selection rescues them.

Critically, the structural metric *saturates*: even frozen rules at 0.58 accuracy realize the full γ -separated code (24/24 clusters). Quantifying this against Corollary 1 ($d=32$, $V=48$): the realized code is

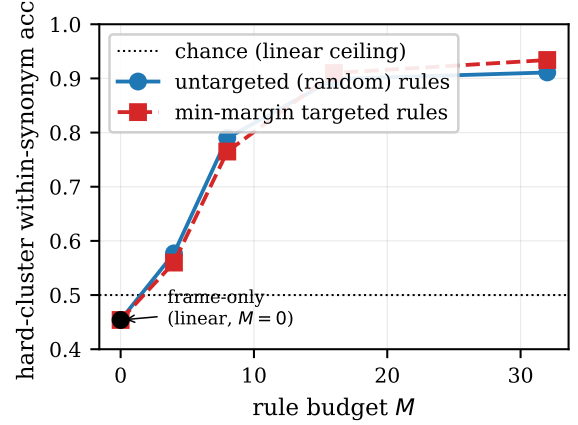


Figure 2: Generation is the lever; targeting is not. A frame-only linear read-out is pinned at chance on the XOR-coded hard clusters ($M=0$); generated rules lift hard-cluster within-synonym accuracy to 0.93. Min-margin targeting (dashed) is statistically indistinguishable from random rules (solid)—SGD already allocates rules to the at-risk facets.

Table 3: The decisive 2 $\times$ 2: training, not selection, is the lever (hard-cluster within-synonym accuracy, held-out, 3 seeds, chance 0.5). The code column is the realized γ -separated code (clusters with both synonyms confidently decoded; max 24).

M	sgd	ambig-frozen	ambig-train	rand-frozen	rand-train
8	0.722	0.532	0.741	0.540	0.750
16	0.859	0.579	0.852	0.632	0.854
32	0.905	0.684	0.891	0.668	0.873
code	24	24	24	24	24

48/48 tokens, while the packing bound is $\sim 10^{59}$ at $\gamma=1$ (and $\sim 10^{51}$ at $\gamma=1.8$). *Capacity is thus slack by ~ 50 – 59 orders of magnitude.* The minimum frame separation inside the code (0.90) further exceeds the effective γ by 17–31 $\times$. So creating new γ -separated high-margin tokens is not the bottleneck; that capacity is realized with enormous slack. The binding constraint is per-example routing: getting the *right* decode for *each* input, which is a training problem, not a capacity or selection one.

5.4 Realization cost is M -dominated—a superposition quantity

We next operationalize TRC. Holding cell capacity fixed (slack $\sim 10^{59}$), we vary the number of independent non-linear routing decisions n_{hard} and the rule budget M (Fig. 3). The natural hypothesis—accuracy is governed by rules-per-decision M/n_{hard} —is *refuted*: accuracy depends on the absolute budget M and essentially not on n_{hard} . At $M=8$ all of $n_{\text{hard}} \in \{4, \dots, 24\}$ reach ~ 0.76 ; at $M=32$ all reach ~ 0.91 . The minimal M to reach 0.85 is ~ 16 – 32 regardless of n_{hard} , so M^*/n_{hard} falls from 4.0 to 0.67: 24 disjoint XOR routings need no more rules than 4.

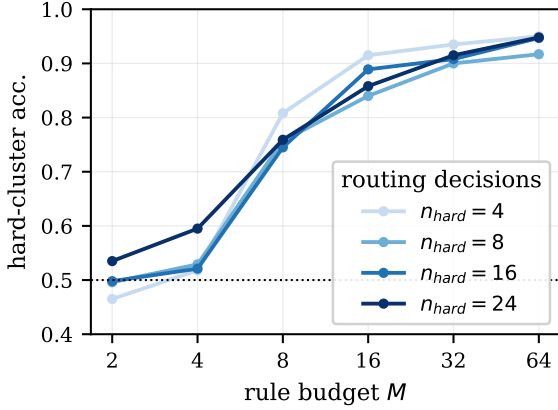


Figure 3: Realization cost is dominated by the rule budget M , not the number of routing decisions n_{hard} : the curves collapse against M , not against M/n_{hard} . The M rules represent the n_{hard} routing features in superposition.

Table 4: Interference is mild and lives in $\mathbb{R}^M$ (held-out, 3 seeds, $d=32$). Coherence μ reaches the Welch floor only once $n_{\text{hard}} > M$; the margin degrades only weakly there.

M	margin ($n=1 \rightarrow 24$)	μ	Welch ($n=24$)	verdict
8	2.10 $\rightarrow$ 1.83	~ 0.28	0.295	mild, only $n > M$
16	2.93 $\rightarrow$ 3.11	~ 0.19	0.147	no degradation

The interpretation is superposition: the M rules represent the n_{hard} routing features amortized across decisions, with accuracy set by how many features M rules can disentangle, not by the decision count. **TRC is therefore a superposition-capacity quantity**, not a per-decision count—the same superposition the broader program studies, now on the routing side.

This reframes the interference term. Probing the margin and the realized coherence μ of the per-cluster routing directions (Table 4) gives three readings. First, there is no clean Welch-degradation cliff: $M=16$ is flat, and $M=8$ declines only $\sim 13\%$. Second, the relevant dimension in Theorem 4 is M , not $\min(M, d)$ —the routing features live in *rule-activation space* $\mathbb{R}^M$, so interference appears precisely when $n_{\text{hard}} > M$, where μ sits at the Welch floor. Third, training packs near-optimally: realized μ tracks the floor without wildly exceeding it, so exceeding M costs only mild margin, not failure. A direct generator-side coherence regularizer confirms the picture—it lowers μ on target but leaves margin and accuracy flat ($|\Delta \text{margin}| \leq 0.04$), even at 2–3 $\times$ over-packing.

5.5 Four nulls, one lesson

Four theory-guided knobs have now been tested against plain end-to-end SGD (Table 5); none beats it. The consistent lever is simply $M \geq$ the effective rank of the routing features, with SGD packing them near the Welch floor on its own. This is also why a coherence $\rightarrow$ margin degradation theorem is not worth formalizing: coherence is real and Welch-floored (Theorem 4), but it is

Table 5: The symmetry closes: four theory-guided interventions, all null vs. SGD.

knob	side	result
frame_reg (decorrelate U)	frame	null (§5.1)
min-margin rule targeting	generator	null (§5.2)
propose-score-select (frozen rules)	generator	null (§5.3)
coh_reg (decorrelate firing)	generator	null (§5.4)

empirically decoupled from the margin. The honest theory boundary is that the *structural packing bounds are provable* (and proved), while the *margin consequence* is mild and not a clean law.

Why this calibrates the real-model results. These nulls are not a dead end; they are the measuring stick for §6. Because SGD packs near the Welch floor whenever it is *trained on a task*, tight routing has a known signature—rank used up, coherence $\sim 1\times$ the floor. Any slack a *pretrained* model shows against that signature (rank left unused, coherence above the floor) is therefore not an optimization failure but a structural fact about a model that was never specialized to a single routing decomposition. The synthetic regime fixes the reference point so that the real-model departures from it are interpretable rather than arbitrary. *In one line: wherever SGD can reach, the theory’s role is to bound and measure the packing, not to improve it.*

6 EXPERIMENTS: REAL MODELS

This is the section the framework is built for. Four claims, each measured on a faithful decompilation of a real model and tagged [MEASURED]:

- (1) **Faithful seam.** The emitted incidences reconstruct the model’s own decode 1.00 of the time—on every model, corpus, and language, and across two architectures (Qwen2.5 and GPT-NeoX/Pythia) (§6.1).
- (2) **Rank concentration on a stable circuit transfers across families.** In both Qwen and Pythia the decode rides a small, consistent, late-layer, MLP-dominated read-out circuit—far fewer effective blocks than n_b (§6.2).
- (3) **Coherence packing is family-dependent.** Qwen routing features sit 2–3 $\times$ above the Welch floor; Pythia packs near it (1.06–1.5 $\times$). At matched dimension, Pythia-410M (1.38 $\times$) is far tighter than Qwen2.5-0.5B (2.01 $\times$)—so near-optimal packing is *not* synthetic-specific (§6.3).
- (4) **Difficulty increases slack via fallback (Qwen).** Harder inputs route through *fewer*, looser blocks, not more—the opposite of the naive expectation (§6.4).

6.1 The fieldrun seam

We validate the structural theory on real models through `fieldrun --pil-dump`. For each natural-text position it emits the per-block DLA incidence matrix $\text{contrib}[b][v] = \langle d_b, U_v \rangle$ over the top- K candidate tokens, where the blocks are the $n_b = 2L+1$ additive residual components. In the PIC framing each block is a rule, its per-candidate contribution an incidence, and the decode logit is their additive sum. The seam is *faithful*: across every model, corpus, and language, the emitted incidences reconstruct the decode

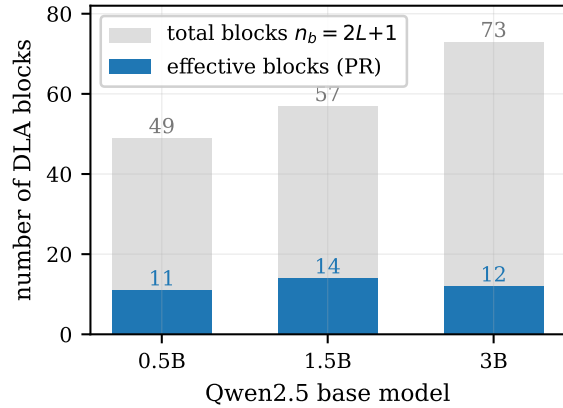


Figure 4: Rank concentration across scale (Qwen2.5 base, prose). The effective number of blocks driving the decode stays $\sim$ constant (~ 12) while the total block count $n_b = 2L+1$ grows, so rank slack grows with scale.

Table 6: Real-model routing across scale (Qwen2.5 base, prose). $PR/n_b \ll 1$ is rank concentration; $\mu/\text{Welch} \gg 1$ is coherence slack.

model	n_b	recon	PR/n_b	μ	μ/Welch	margin
0.5B	49	1.00	11/49	0.268	2.01	1.24
1.5B	57	1.00	14/57	0.261	2.18	1.27
3B	73	1.00	12/73	0.339	3.62	1.30

$\arg \max$ 1.00 of the time and the recomputed margin equals the model’s exactly. (We analyze the model’s *own* decode— $\arg \max$ vs. runner-up—not correctness.) Models: Qwen2.5 base 0.5B/1.5B/3B and 0.5B-Instruct/Coder ($n_b = 49/57/73$), and the Pythia scaling suite [3] 70M/160M/410M/1B ($n_b = 13/25/49/33$), a second pre-training corpus (the Pile) and architecture (GPT-NeoX). The seam reconstructs the decode 1.00 of the time on *every* one, so all comparisons below are between faithful decompilations.

6.2 Routing is rank-concentrated on a stable circuit

The decode is driven by a small effective number of blocks—the source participation ratio over per-block contributions—far below n_b (Fig. 4, Table 6). Strikingly, within Qwen2.5 the effective count is $\sim$ constant (~ 12) *regardless of scale*, so as n_b grows the rank *slack* grows (PR/n_b 0.22 $\rightarrow$ 0.16). This is the empirical face of Theorem 3: the model uses only a fraction of the available routing dimensions, and within this family that fraction shrinks with scale.

Moreover the concentration is not a different random subset each token: the global participation ratio over the mean per-block contribution tracks the per-position ratio (e.g. 14.8 vs. 11.4 for 0.5B), and the global top-12 blocks capture 73–78% of per-position decode mass (Table 7). The effective routing is a *stable, identifiable circuit*, concentrated in late layers (78–85% of mass) and dominated by MLP

Table 7: The effective blocks form a consistent late-MLP circuit (Qwen2.5 base, prose). Global $PR \approx$ per-position PR , and the global top-12 blocks carry most of the per-position decode mass.

model	per-pos PR	global PR	top-12 mass	late	MLP
0.5B	11.4	14.8	75%	78%	76%
1.5B	14.0	17.9	73%	81%	73%
3B	11.8	15.0	76%	85%	78%

Table 8: Real-model routing across scale on Pythia (Pile, GPT-NeoX; prose). Rank concentration ($PR/n_b \ll 1$) transfers, but coherence packing μ/Welch is *near-optimal* (1.06–1.5 $\times$), unlike Qwen (Table 6).

model	n_b	recon	PR/n_b	μ	μ/Welch	margin
70M	13	1.00	2.9/13	0.289	1.06	0.98
160M	25	1.00	2.6/25	0.269	1.39	1.11
410M	49	1.00	20.7/49	0.184	1.38	1.18
1B	33	1.00	13.9/33	0.249	1.50	1.21

blocks (73–84%). Rank concentration is thus a structural property—a small late-MLP read-out circuit—not merely a per-token statistic.

The same analysis on the Pythia suite (Table 8)—a second pre-training corpus (the Pile) and architecture (GPT-NeoX)—confirms rank concentration in a different family: $PR \ll n_b$ throughout, again on a *consistent* circuit (global $PR \approx$ per-position PR), late- and MLP-leaning. The scale *trend* differs from Qwen, however. The smallest Pythia models concentrate hardest— ~ 3 of 13–25 blocks, with a large embedding share (30–37% of decode mass) and 97% of mass in the late third—and the effective count *rises* with size to ~ 14 –21. So the cross-family invariant is rank concentration itself, not Qwen’s near-constant ~ 12 . The axis on which the families genuinely part ways is coherence (the μ/Welch column), which we take up next.

6.3 Coherence packing is family-dependent

Treat each position’s decode-vs-runner-up direction in block space as a routing feature, and measure their mutual coherence μ against the Welch floor $\sqrt{(N - n_b)/(n_b(N - 1))}$ (Theorem 4; Fig. 5). Here the two families diverge, and the divergence is the single sharpest signal in the real-model data. *Qwen leaves substantial slack*— μ/Welch sits 2–3.6 $\times$ above the floor in every condition ($\mu \approx 0.26$ –0.41). *Pythia does not*: it packs near-optimally, $\mu/\text{Welch} = 1.06$ –1.5 $\times$, hugging the same floor a task-specialized synthetic learner reaches.

The cleanest comparison controls the dimension. At $n_b=49$ and $N \approx 380$ (identical routing-feature geometry), Pythia-410M packs at $1.38\times$ Welch ($\mu=0.18$) while Qwen2.5-0.5B sits at $2.01\times$ ($\mu=0.27$)—a genuine difference in realized coherence, not a floor artifact. So near-Welch-optimal packing, which we previously observed only in task-specialized SGD (§5.4), *does* occur in a general pretrained family: it is a property of the training recipe (corpus, optimizer, regularization), not a universal consequence of pretraining. The structural bound holds for both families: the Welch floor is real and never violated. What differs—and what the diagnostic measures—is

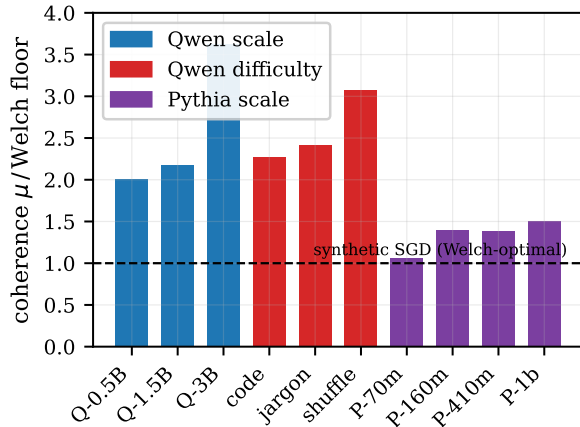


Figure 5: Coherence relative to the Welch floor, by family. A task-specialized synthetic learner packs to $\sim 1\times$ (dashed line). Qwen2.5 (scale and difficulty conditions) sits 2–3.6 $\times$ above it; Pythia hugs it (1.06–1.5 $\times$). Near-optimal packing is thus family-dependent, not synthetic-specific.

Table 9: Difficulty dial (Qwen2.5-0.5B-Instruct). Harder inputs (lower margin) produce *more* slack: fewer effective blocks, looser packing.

corpus	margin ($\downarrow$ harder)	PR/ n_b	μ	μ/Welch
prose (easy)	1.24	11/49	0.268	2.01
code	2.56	12/49	0.288	2.27
rare-word jargon	1.74	9/49	0.305	2.41
shuffled (hard)	0.48	9/49	0.407	3.07

how much slack each family leaves above it, and that separates Qwen from Pythia cleanly (Fig. 5). (The smallest Pythia models have small n_b and thus a high floor with little room to be loose, so the $n_b=49$ comparison is the controlled one.)

6.4 Difficulty increases slack via fallback

Is the slack an easy-corpus artifact? We varied input difficulty (using the model’s own decode margin as the proxy). The result is the *opposite* of the naive expectation (Table 9). The genuinely hard corpus (shuffled word order, margin 0.48) gives the *most* slack: fewer effective blocks (9/49) and looser packing ($\mu=0.41$). The mechanism is fallback—a confused model leans on a few generic high-prior blocks. (Code is a red herring: its rigid syntax makes the model *more* confident, margin 2.56; rare-word jargon is mostly retrieval-style subword completion.) So the rank slack and looseness are intrinsic and grow with difficulty.

Multilingual robustness. The picture is qualitatively robust across five languages (English, Chinese, French, German, Spanish): all reconstruct 1.00, routing stays rank-concentrated (~ 7 –11 of 49 blocks), and coherence stays in the Qwen-typical loose band, slightly looser for the less-fluent languages—consistent with the difficulty finding. The per-language numbers are deferred to Appendix A.

7 DISCUSSION

The near-term value of PIC and TRC is *diagnostic*, not algorithmic: measuring rank concentration, identifying the read-out circuit, quantifying coherence slack, and detecting difficulty-driven fallback—quantities a practitioner can read off a frozen model with a faithful DLA seam, and which behave lawfully across scale, difficulty, and language. The rest of this section explains why the framework lands on the diagnostic side of that line, and why that is the right place for it.

Combining the proved structural bounds (Theorems 2–4) with the synthetic and real measurements yields a coherent picture. *Cell capacity is not the operative limit*: it is slack by ~ 50 –59 orders of magnitude (§5.3), and both real-model families confirm large rank slack (§6). The operative quantity is *TRC*, and it is a superposition quantity—how many routing features a budget of M generators can pack with tolerable interference (§5.4). On the synthetic substrate, plain SGD already performs that packing near the Welch floor, which is exactly why the four theory-guided interventions are null: there is no slack for them to recover. The contribution of the theory there is not a better optimizer but the *ceiling and the coordinates*—what is reachable, and in which basis.

Real models then sort into two regimes the diagnostic makes visible. *Rank concentration transfers universally*: both Qwen and Pythia route through a small, stable, late-MLP circuit, the empirical face of the rank bound (Theorem 3). *Coherence packing does not*: Qwen leaves 2–3 $\times$ slack above the Welch floor, whereas Pythia packs near it (1.06–1.5 $\times$), as tightly as task-specialized synthetic SGD. We had initially read near-optimal packing as a signature of task specialization that general pretraining would not pay for; the Pythia result refutes that and is the more interesting finding. Whether a family packs its routing features tightly is a property of the *training recipe*—corpus, optimizer, regularization—not a universal consequence of pretraining, and not something accuracy or loss curves reveal. PIC is the instrument that surfaces it: a single scalar, μ/Welch , that cleanly separates two model families trained to comparable competence.

8 RELATED WORK

8.1 Tropical geometry and neural networks

Zhang et al. [20] established the foundational connection between feedforward ReLU networks and tropical geometry, showing that such networks compute tropical rational maps, with decision boundaries corresponding to tropical hypersurfaces and linear regions to vertices of associated Newton polytopes. This has been extended to real tropical geometry [4] and applied to decision-boundary analysis [2] and linear-region counting [11, 17]. *All of this prior work addresses the input–output map of feedforward classifiers*. We instead apply tropical geometry to the *routing and read-out behavior of decoder-only transformers*, via direct logit attribution. The relevant geometric object is then the Laguerre power diagram of the affine logit functions, and the central structural result is a γ -code packing *count* on confidently decodable tokens (Corollary 1)—the decision-side analogue of the Welch coherence bound rather than a region count.

8.2 Superposition in neural networks

Our coherence analysis provides a routing-side instance of superposition. A network can represent more features than it has dimensions only by accepting interference, and that interference is bounded below by coherence measures such as the Welch bound [19]. This is the lens of toy-model and complexity-theoretic work on superposition [1, 7], which has focused on *hidden representations*. We extend the same accounting to the *routing* side. When the number of routing features n exceeds the effective dimension M of the rule-activation space, their coherence is Welch-floored (Theorem 4). We further show empirically that gradient descent attains near-optimal packing ($\sim 1\times$ the floor) on task-specific substrates, whereas general pretrained models leave measurable slack ($2\text{--}3\times$ the floor) on naturally occurring routing features (§5.4–6). Routing, in other words, is superposition with M as the packing dimension.

8.3 Low-rank and routing in transformers

Efficiency-oriented work explores low-rank and sparse approximations to attention [6] to reduce computational cost. Our contribution is complementary and orthogonal: we *measure* naturally occurring rank concentration in the routing behavior of pretrained models as an emergent structural property, rather than imposing low-rank structure for efficiency.

8.4 Mechanistic interpretability and incidence calculus

Our analysis builds on direct logit attribution and the residual-stream view of transformer circuits [8, 12]. On top of that additive decomposition we introduce a geometric and logical semantics (incidence calculus plus tropical geometry) together with a capacity theory (Tropical Realization Complexity), and we modernize classical incidence calculus [5, 9] for the inner-product structure of transformer residuals—replacing world-sets with signed incidences and structural overlap with a Gram kernel.

9 LIMITATIONS AND THREATS TO VALIDITY

The real-model analysis rests on one design choice—“routing feature = decode-vs-runner-up direction in block space”—which is reasonable but not unique; alternative definitions could shift the coherence numbers. The DLA blocks are fieldrun’s decomposition, not the only circuit basis. Non-English samples are small ($N=96\text{--}127$), so per-language coherence is noisy (the qualitative trend is the signal). The synthetic substrate is deliberately stylized (planted clusters, XOR coding); its value is a controllable ground truth, not ecological realism. The machine-checked results (Theorems 1–4) are verified in the i-orca corpus [15]; the packing constant in Corollary 1 is the standard covering-number corollary of Theorem 2. Finally, we cover two families (Qwen2.5 and Pythia): they already show that coherence packing is family-dependent rather than universal, so single-family conclusions about packing should not be generalized—identifying what in a training recipe sets the packing tightness, across more families and corpora, remains open.

10 CONCLUSION AND FUTURE WORK

We presented Projective Incidence Calculus as a geometric and logical account of transformer decision making, together with Tropical Realization Complexity as a diagnostic for routing cost. We proved the structural backbone—exact decode recovery, a decode-separation/packing bound, generator rank, and a routing Welch floor—in a machine-checked corpus. A deliberate set of four null interventions then showed that the framework’s value is diagnostic rather than algorithmic: capacity is hugely slack, and SGD already packs near the floor. Across two model families, real-model analysis then shows that rank-concentrated routing on a stable late-MLP circuit is universal, whereas coherence packing is *family-dependent*—loose in Qwen ($2\text{--}3\times$ Welch), near-optimal in Pythia ($1.06\text{--}1.5\times$).

Future directions include: practical estimators for TRC on large models; whether narrow fine-tuning tightens packing toward the synthetic ($\sim 1\times$) regime; hybrid generation that activates only when diagnostics indicate rank or interference pressure; and—prompted by the Qwen/Pythia split—identifying which ingredients of a training recipe set routing-packing tightness, across more families and corpora. All code, formal certificates, and results are available in the fieldrun, i-orca, and pil repositories.

REFERENCES

- [1] Micah Adler and Nir Shavit. 2024. On the Complexity of Neural Computation in Superposition. *arXiv preprint arXiv:2409.15318* (2024).
- [2] Motasem Alfarra, Adel Bibi, Hasan Hammoud, Mohamed Gaafar, and Bernard Ghanem. 2023. On the Decision Boundaries of Neural Networks: A Tropical Geometry Perspective. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 4 (2023).
- [3] Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shrivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. 2023. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- [4] Marie-Charlotte Brandenburg, Georg Loho, and Guido Montúfar. 2024. The Real Tropical Geometry of Neural Networks. *arXiv preprint arXiv:2403.11871* (2024).
- [5] Alan Bundy. 1985. Incidence Calculus: A Mechanism for Probabilistic Reasoning. *Journal of Automated Reasoning* 1, 3 (1985), 263–283.
- [6] Beidi Chen, Tri Dao, Eric Winsor, Zhao Song, Atri Rudra, and Christopher Ré. 2021. Scatterbrain: Unifying Sparse and Low-rank Attention. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [7] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. 2022. Toy Models of Superposition. *Transformer Circuits Thread*.
- [8] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, et al. 2021. A Mathematical Framework for Transformer Circuits. *Transformer Circuits Thread*.
- [9] Weiru Liu and Alan Bundy. 1996. A Comprehensive Comparison Between Generalized Incidence Calculus and the Dempster–Shafer Theory of Evidence. *International Journal of Human-Computer Studies* 45, 1 (1996), 1–38.
- [10] Diane MacLagan and Bernd Sturmfels. 2015. *Introduction to Tropical Geometry*. Graduate Studies in Mathematics, Vol. 161. American Mathematical Society.
- [11] Guido Montúfar, Razvan Pascanu, Kyunghyun Cho, and Yoshua Bengio. 2014. On the Number of Linear Regions of Deep Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [12] nostalgebraist. 2020. Interpreting GPT: The Logit Lens. *AI Alignment Forum*.
- [13] Qwen Team. 2025. Qwen2.5 Technical Report.
- [14] J. Allan Scott. 2026. fieldrun: Projective Incidence Calculus probing infrastructure for transformer language models. Software, <https://github.com/jascal/fieldrun>. SOFTWARE.
- [15] J. Allan Scott. 2026. i-orca: a machine-checked formalization corpus for incidence and tropical decode certificates. Formal corpus, <https://github.com/jascal/i-orca>. SOFTWARE.
- [16] J. Allan Scott. 2026. pil: Projective Incidence Learning — learning dynamics for Projective Incidence Calculus. Software, <https://github.com/jascal/pil>. SOFTWARE.

- [17] Thiago Serra, Christian Tjandraatmadja, and Srikumar Ramalingam. 2018. Bounding and Counting Linear Regions of Deep Neural Networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML) (PMLR, Vol. 80)*.
- [18] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [19] Lloyd R. Welch. 1974. Lower Bounds on the Maximum Cross Correlation of Signals. *IEEE Transactions on Information Theory* 20, 3 (1974), 397–399.
- [20] Liwen Zhang, Gregory Naitzat, and Lek-Heng Lim. 2018. Tropical Geometry of Deep Neural Networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML) (PMLR, Vol. 80)*. 5824–5832.

A MULTILINGUAL ROUTING DETAIL

Table 10 reports the per-language routing numbers for Qwen2.5-0.5B-Instruct on matched content, supporting the robustness claim in §6.4. All five languages reconstruct the decode 1.00 of the time, route through ~ 7 –11 of 49 blocks, and show $\mu \approx 0.27$ –0.36. Non-English is slightly *more* concentrated and looser, consistent with

the difficulty finding (marginally lower fluency $\rightarrow$ more fallback); Chinese matches English coherence ($\mu=0.269$) despite the script. Non-English N is small (96–127), so μ is noisier; the qualitative picture is the signal.

Table 10: Multilingual routing (Qwen2.5-0.5B-Instruct, matched content, $n_b = 49$).

language	PR/ n_b	μ	margin
en	11/49	0.268	1.24
zh	8/49	0.269	1.12
fr	8/49	0.329	1.89
de	8/49	0.357	1.54
es	7/49	0.343	1.99